

Musa SAHIN

SENIOR SOFTWARE ENGINEER

✉ sahinmusa@gmail.com ☎ +90 505 383 61 43 📍 Istanbul, Turkiye

🌐 LinkedIn 🔄 GitHub 📄 StackOverflow 📺 Youtube

PROFESSIONAL SUMMARY

Software Engineer with 9 years of experience. Primary focus on **Flutter (7 years)** and **NestJS (2 years)**, with additional practical experience in native **iOS (Swift)**, **Android (Kotlin)**, and **React Native**. Experienced in building scalable architectures, implementing mobile security standards, and establishing automated testing workflows through **TDD** & Clean Architecture. Specialized in **FinTech**, **Geomatics**, **E-commerce**, **Education**, and **Gaming** domains. Conducted thesis research on **NLP** (Natural Language Processing). Implements algorithms directly from research papers (as seen in **Guardrail**) to build custom AI agents and autonomous systems.

WORK EXPERIENCE

Senior Mobile Developer Orion Innovation (Edison, NJ - USA Remote)	Oct 2024 - Present
- Developing HuntStand, a cross-platform (Mobile & Web) application for Hunting and Land Management using Flutter. - Utilizing Generative AI tools to optimize code quality, automate refactoring, and accelerate feature implementation. - Updated the internal MapSDK to improve geospatial rendering performance across multiple products. - Implemented Test-Driven Development (TDD) workflows, addressing technical debt in legacy modules. - Conducting code reviews and mentoring team members on modular architecture and state management.	
Software Developer Engineerica Inc (Oviedo, FL - USA Remote)	Jul 2023 - Oct 2025
- Developed and maintained AccuBus, a mobile application for monitoring state school bus logistics. - Integrated AI-assisted workflows in the development process to enhance problem-solving speed and efficient code documentation. - Rewrote the iAccu attendance mobile and kiosk application in Flutter for cross-platform support. - Contributed to the Accudemia academic center management system using native Swift and Kotlin. - Developed AccuKiosk, a lightweight version of the attendance tracking system. - Built AccuClass, a class management and scheduling application. - Implemented My Conference Planner for event scheduling and resource management.	
Software Developer & Product Manager Optimus Bilişim (Istanbul, Turkiye)	Jan 2024 - Aug 2024
- Architected white-label stock trading applications and SDKs used by multiple financial institutions (e.g., Meksa Mobile). - Refactored legacy codebases to improve architectural modularity and maintainability. - Implemented security measures to meet penetration testing requirements for financial software. - Managed the complete application rewrite to resolve performance issues.	
Software Developer & Team Lead Innovance / Oyak Yatırım (Istanbul, Turkiye)	Mar 2023 - Jan 2024
- Led the mobile development team for the Oyak Finance Trader application. - Established architectural standards and managed the release pipeline for the Flutter codebase.	
Full Stack Software Developer Nakit Yok (Ukraine Antalya, Turkiye)	Mar 2023 - Mar 2024
- Developed NOSH, a food ordering and payment mobile application using Flutter. - Built the backend infrastructure using NestJS and PostgreSQL.	
Software Developer Aymar Tech (Istanbul, Turkiye)	Jul 2022 - Mar 2023
- Led the implementation of "Untangible", an NFT marketplace and Web3-integrated game platform using React Native and Flutter. - Integrated Web3 wallets and smart contract interactions within both React Native and Flutter environments.	
Flutter Engineer United Software / Wio Bank (Ukraine UAE Remote)	Apr 2022 - Feb 2023
- Contributed to developing both the Web and Mobile versions of Wio Business, a digital banking application. - Collaborated with distributed teams to ensure banking compliance and security.	
Principal Mobile Specialist Etiya Information Technologies (Istanbul, Turkiye)	Feb 2022 - Apr 2022
- Led development of the Trumore Super App (Official App for Togg Car) using Flutter. - Implemented features to meet automotive industry performance standards.	
Software Developer & Team Lead DOP (Istanbul, Turkiye)	Oct 2021 - Feb 2022
- Led mobile development team for Siparişim and Restoran İşim Mobil applications. - Implemented micro-frontend architecture to improve scalability across mobile platforms. - Mentored junior developers on Flutter best practices and architectural patterns. - Contributed to backend development using Python (FastAPI) and NestJS.	
Software Developer Internative Software (Istanbul, Turkiye)	Mar 2021 - Oct 2021
- Developed Ali4Baby, a parents and baby assistant application using Flutter. - Built Sodio, a social radio application using custom Flutter UI components. - Implemented interactive mobile UI designs focused on user engagement.	
Software Developer S'aide Financial Technologies (Istanbul, Turkiye)	Apr 2020 - Feb 2021
- Developed S'aide, a student assistant and payment application using Flutter. - Integrated payment gateway APIs for secure transaction processing. - Implemented security measures to pass penetration testing requirements for financial applications. - Built the initial backend infrastructure using NestJS and MongoDB.	
Full Stack Software Developer Freelance	Jun 2017 - Apr 2020
- Developed web and mobile applications for clients using Ionic, React, React Native, and Flutter. - Delivered custom solutions for small to mid-sized projects across various industries.	

PERSONAL PROJECTS

Flash Game Engine Open Source
- Created a lightweight 2.5D game engine for Flutter (written in C++ & Dart) featuring declarative widgets, physics simulation, and spatial audio. - Successfully used to develop 10+ high-quality demo games across mobile platforms. - Designed a custom entity-component system (ECS) optimized for Dart and Flutter's rendering pipeline.
Guardrail AI Code Intelligence & Security Platform
Platform Overview & Impact: Developed an advanced AI Code Intelligence platform to automate security and code quality checks. Successfully pilot tested at Orion Innovation, demonstrating capabilities in reducing technical debt and enhancing review workflows. Agent Development & Orchestration: Built a custom, recursive RLM framework (arXiv:2512.24601) with autonomous planning (map-reduce tasks) and self-corrections via verification loops. Designed a Multi-Agent System (Explorer, Analyst, Critic) that executes code safely within a Python REPL Sandbox. RAG & Knowledge Systems: Architected a Hybrid RAG system combining Semantic Search (ChromaDB) with Knowledge Graphs (KuzuDB). Mapped code structures (AST, function calls) to answer complex "Blast Radius" queries with 99% accuracy. LLMOps & Local Deployment: Engineered a privacy-first Local LLM Pipeline using Ollama (SLM models). Implemented Cost Optimization strategies with model routing and L1/L2 caching, ensuring efficient enterprise deployment. Dataset Generation & Fine-Tuning: Implemented a Human-in-the-Loop mechanism that captures developer feedback and stores it in Graph & Vector DBs to build high-quality datasets for future LoRA Fine-Tuning. Enterprise Integration: Designed agents capable of interacting with corporate ecosystems (Jira, GitHub/GitLab APIs) for automated issue tracking and CI/CD analysis, effectively bridging the gap between AI reasoning and DevOps workflows.

TECHNICAL SKILLS

MOBILE DEVELOPMENT
Flutter (7 years) Dart iOS (Swift) Android (Kotlin) React Native BLoC / Cubit
BACKEND & API
NestJS NodeJS Python (FastAPI) TypeScript RestAPI GraphQL gRPC
WebSockets RabbitMQ Redis
DATABASE
Graph Databases (KuzuDB, Neo4j) Vector Search (ChromaDB) PostgreSQL MongoDB
Firebase
CLOUD & DEVOPS
Docker CI/CD (Gitlab, Github Actions) LLMOps Pipelines Model Serving (vLLM)
AI & LLM ENGINEERING
Recursive RLM (Paper-Based) Multi-Agent Systems Hybrid RAG (Graph + Vector) Local SLMs
AI Safety & Sandbox Cost & Model Routing
METHODOLOGIES & CONCEPTS
TDD Clean Architecture Microservices SOLID Design Patterns Agile / Scrum OOP

EDUCATION

BS, Information Systems and Technologies Yeditepe University - Istanbul, Turkiye	2021
BS, Management Information Systems Anadolu University - Eskisehir, Turkiye	2020
BS, Physics (Double Major) Yeditepe University - Istanbul, Turkiye	2021
MEd, Science Teacher Education Marmara University - Istanbul, Turkiye	2020
Thesis: Utilizing NLP (Natural Language Processing) for Assessing Open-Ended Examinations	
B.E. in Chemistry Engineering Marmara University - Istanbul, Turkiye	2015
BSc, Chemistry Marmara University - Istanbul, Turkiye	

LANGUAGES

Turkish: Native	English: Professional Proficiency
-----------------	-----------------------------------

